

NORTHEASTERN UNIVERSITY

EXPLORATORY DATA ANALYSIS (EDA) PROJECT

FUTURE OF WORK IN AI ERA

Research Questions:

- 1. Which job roles carry the highest automation risk, and does risk level vary meaningfully across industries?*
- 2. Which roles face the most urgent need for reskilling, and what does that pressure look like compared to their current skill gap?*

Source: Kaggle – ai_job_replacement_2020_2026_v2.csv

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I – INTRODUCTION

1.1. Dataset Overview

This report presents an exploratory data analysis of the AI Job Replacement Dataset (2020 - 2026). This is a synthetic dataset containing 15,000 records across 20 variables. The dataset simulates workforce conditions across 10 job roles, 8 industries, and 9 countries over six years from 2020 to 2026.

1.2. Data Dictionary

The dataset contains 16 numerical and 4 categorical variables. Key fields are described below:

Variable	Type	Description
job_id	Numerical	Unique number for each row
job_role	Categorical	10 roles: Truck Driver, Customer Support Rep, Accountant, Financial Analyst, HR Manager, Marketing Specialist, Mechanical Engineer, Teacher, Data Analyst, Software Engineer.
industry	Categorical	8 industries: Healthcare, Retail, Manufacturing, Finance, Education, Energy, Technology, Transportation.
country	Categorical	9 countries: Japan, USA, Canada, Australia, Singapore, Brazil, Germany, UK, India.
year	Numerical	From 2020 to 2026
automation_risk_percent	Numerical	Probability (%) that job could be automated
ai_replacement_score	Numerical	Score (0-100) for likelihood of AI replacing the role
skill_gap_index	Numerical	Gap between current and required skills in AI era
salary_before_usd	Numerical	Average salary before AI adoption (USD)
salary_after_usd	Numerical	Estimated salary after AI adoption (USD)
salary_change_percent	Numerical	Percentage change in salary due to AI adoption
skill_demand_growth_percent	Numerical	Projects growth in demand for new skills related to the job role

remote_feasibility_score	Numerical	Score indicating how feasible it is for the job role to be performed remotely using digital technologies
ai_adoption_level	Numerical	Level of AI adoption in the industry/role (0-100)
education_requirement_level	Numerical	Level of education: High school, Bachelor, etc.
automation_risk_category	Categorical	Level of automation risk: Low, Medium, High
skill_transition_pressure	Numerical	Level of pressure on workers to transition to new skills
wage_volatility_index	Numerical	Fluctuating wages over time
reskilling_urgency_score	Numerical	Urgency score for workers needing retraining
ai_disruption_intensity	Numerical	Overall level of disruption AI

1.3. Research Questions

Artificial intelligence (AI) is reshaping labor markets at an accelerating pace. Understanding which job roles face the highest replacement risk, and which roles need the most urgent reskilling, is critical for policymakers, educators, and labor forces. This analysis addresses two primary questions:

- Which job roles carry the highest automation risk, and does risk level vary meaningfully across industries?
- Which roles face the most urgent need for reskilling, and what does that pressure look like compared to their current skill gap?

1.4. Data Cleaning and Validation

The dataset is synthetic, hence it required minimal cleaning. A full audit produced the following results:

Table 1. Data profile report

Check	Result	Action Taken
Shape	15,000 rows x 20 columns	None Required
Missing Values	0 / 300,000 data points	None Required
Duplicates	0	None Required

Outlier: <i>ai_replacement_score</i> > 100	94 records	Capped at 100 → Remove
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Domain violation handling: 94 records in the *ai_replacement_score* column contained values between 100.09 and 113.07, exceeding the valid upper bound of 100. These are not statistical outliers in the IQR sense. They are domain constraint violations: values that are structurally impossible given the column's definition.

Three approaches were considered:

- Cap at 100: corrects the value but silently overwrites the original error, making auditing harder.
- Remove the 94 rows: produces a fully valid, auditable dataset $n = 14,906$ (before $n = 15,000$).

Since 0.63% is negligible and the removals are concentrated in two roles: Truck Driver ($n=52$) and Customer Support Rep ($n=42$). This approach does not distort group-level analysis.

The absence of missing values and duplicates across 300,000 data points, combined with near-perfectly uniform numerical distributions discovered, confirms the synthetic origin of the dataset.

II – DATA ANALYSIS

2.1. Five-Number Summary

Figure 1. presents the five-number summary for six representative numerical variables, grouped by risk, salary, and urgency.

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FIVE-NUMBER SUMMARY
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	<i>automation_risk_percent</i>	<i>ai_replacement_score</i>	<i>salary_before_usd</i>	<i>salary_after_usd</i>	<i>salary_change_percent</i>	<i>reskilling_urgency_score</i>
Min	5.00	4.01	30003.69	19022.67	-38.37	2.46
Q1	28.79	28.36	60127.23	59223.46	-6.64	26.98
Median	46.24	45.68	89533.05	88787.33	0.15	35.87
Q3	63.60	62.71	119823.84	118432.97	6.69	44.70
Max	94.98	113.07	149984.06	191961.21	36.92	71.58

Figure 1. Five-Number Summary from Jupyter Notebook

The *automation_risk_percent* spans nearly the full 0-100 range with a median of 46.2%, indicating no single risk tier dominates. The *salary_change_percent* has median near zero +0.15%, but the IQR of $\pm 6.7\%$ and extreme tails from -38% to +37%, which reveal significant polarization beneath the aggregate.

2.2. Outlier Detection

The Five-Number Summary revealed the boundaries of each distribution, but not whether extreme values are statistically unusual. Outlier detection using the IQR method formalizes this: any value below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ is flagged as an outlier. This matters because outliers can distort the mean, inflate standard deviation, and pull regression lines. The IQR method is preferred over Z-score here because it does not assume normality, which is appropriate given our mixed distributions.

The table is well-structured and easy to read, with all the necessary columns: *automation_risk_percent*, *ai_replacement_score*, *salary_before_usd*, *salary_after_usd*, *salary_change_percent*, and *reskilling_urgency_score*. And criteria values: IQR, Lower Fence, Upper Fence, Outlier Count, and Outlier %) clearly presented. The Figure 2.:

IQR OUTLIER DETECTION						
Column	IQR	Lower Fence	Upper Fence	Outlier Count	Outlier %	
automation_risk_percent	34.81	-23.43	115.82	0	0.00	
ai_replacement_score	34.35	-23.17	114.24	0	0.00	
salary_before_usd	59696.61	-29417.70	209368.76	0	0.00	
salary_after_usd	59209.51	-29590.81	207247.23	0	0.00	
salary_change_percent	13.33	-26.64	26.68	114	0.76	
reskilling_urgency_score	17.72	0.41	71.27	2	0.01	

Figure 2. IQR Outlier Detection

- The most critical observation is that *ai_replacement_score* has an Upper Fence of 114.24, meaning the 94 records exceeding 100 are not statistical outliers by IQR definition.
- *reskilling_urgency_score*: only 2 outliers (0.01%), safe to leave as is.
- *salary_before/after_usd*: clean but notice negative fences (-\$29K) are mathematically correct but nonsensical for salary. After checking, the dataset has no actual negative salaries, so it's fine to practice.

Figure 3 presents the frequency distribution and the boxplot to detect outlier for the four categorical variables. The dataset is well-balanced across job roles and industries, reflecting its synthetic design.

IQR Outlier Detection — Key Variables

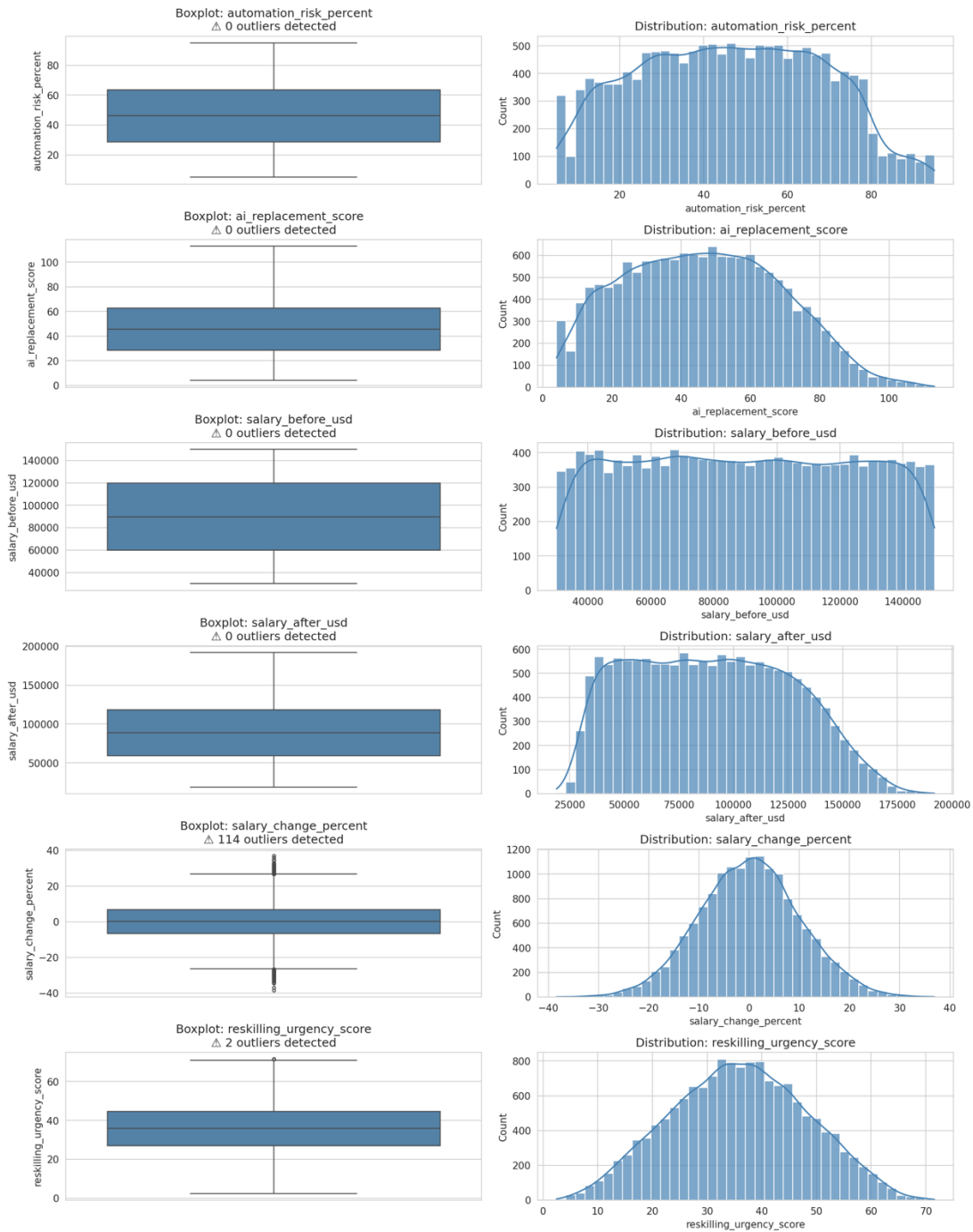


Figure 3. Outlier Boxplot and Distribution Histogram

2.4. Research Question 1: Automation Risk by Job Role and Industry

All statistics below are computed on the cleaned dataset (df_clean , $n = 14,906$).

2.4.1. By Job Role

To answer which job roles carry the highest automation risk, mean *automation_risk_percent* was computed per role and ranked in descending order by horizontal bar chart.

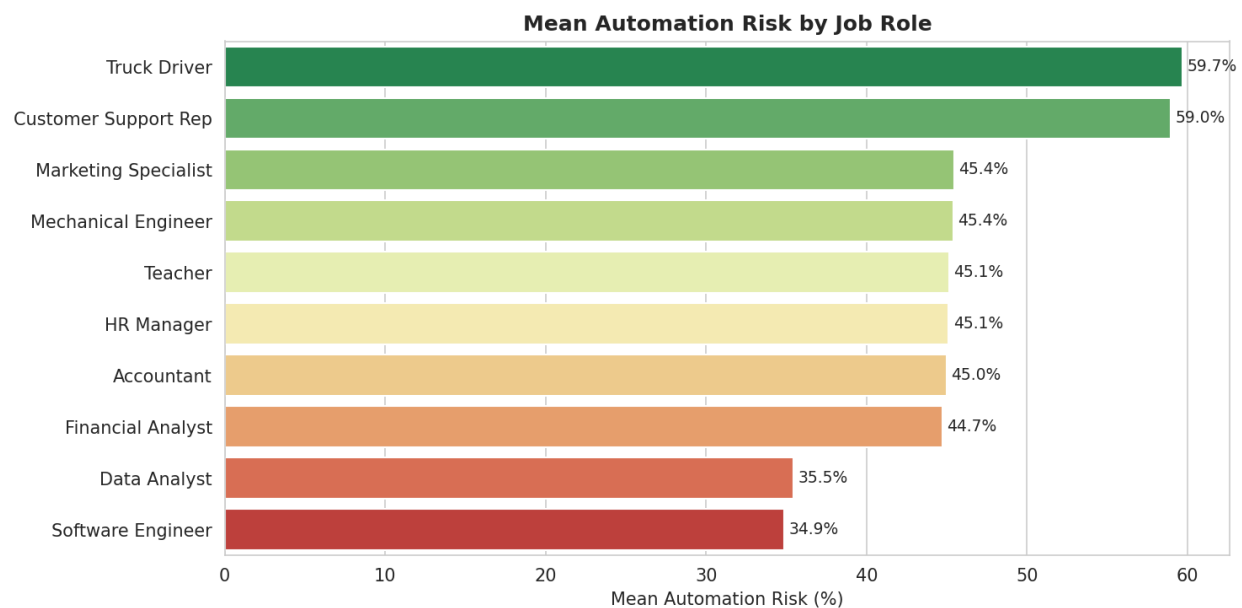


Figure 4. Mean Automation Risk by Job Role

Key takeaways:

- Truck Driver (59.7%) and Customer Support Rep (59.0%) are the highest-risk roles, both are repetitive, rule-based jobs with limited need for creative judgment.
- The middle cluster (Marketing Specialist, Mechanical Engineer, Teacher, HR Manager, Accountant, Financial Analyst) all sit tightly between 44.7 - 45.4%, suggesting moderate and roughly equal exposure.
- Data Analyst (35.5%) and Software Engineer (34.9%) are the lowest risk, somewhat counterintuitive, but reflects that these roles require complex problem-solving, adaptability, and judgment that's harder to automate fully.

2.4.2. By Job Role and Industry

To answer which job roles carry the highest automation risk across the industries, a heatmap cross-tabulating 10 job roles across 8 industries, with Job Role on the vertical axis, Industry on the horizontal axis, and Automation Risk (%) represented by the color scale on the right.

Note: Darker red = higher risk; Darker green = lower risk

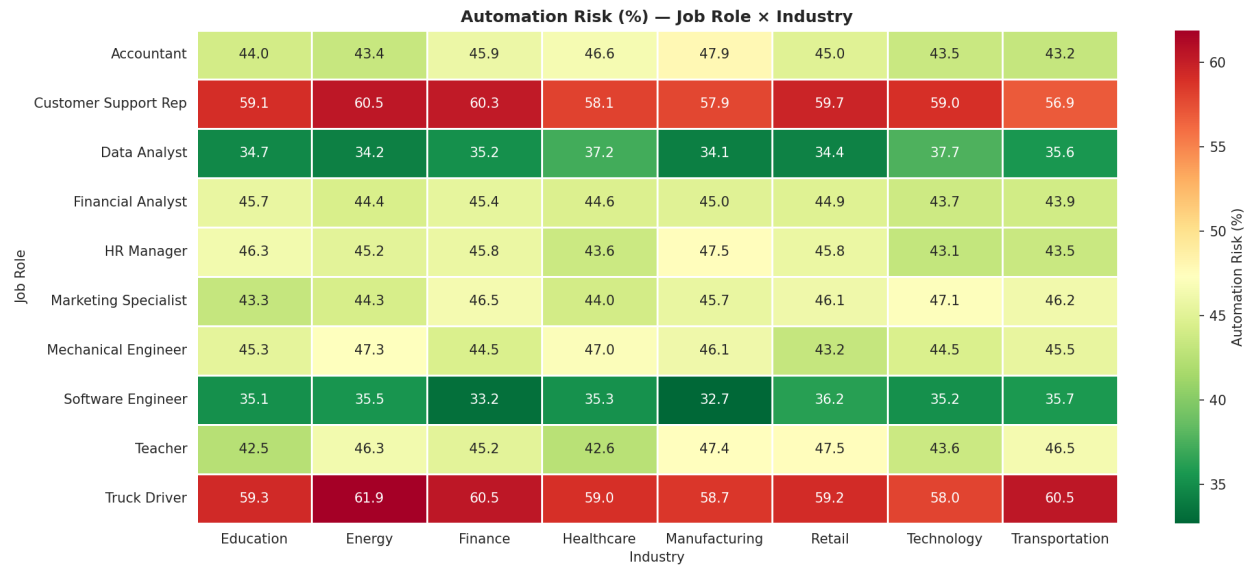


Figure 5. Automation Risk (%) by Job Role × Industry

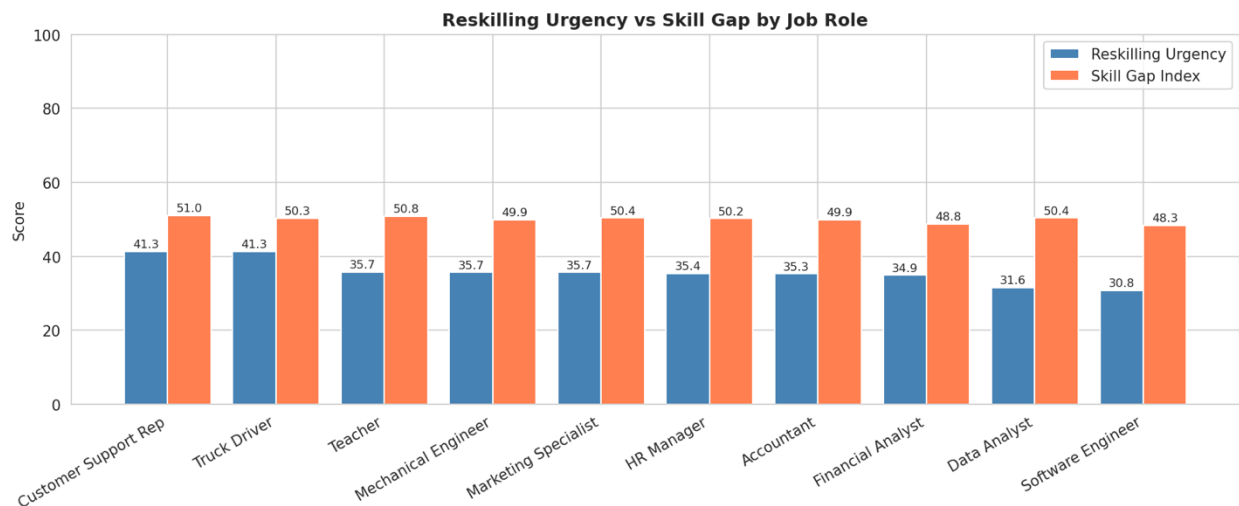
Key takeaways:

- Risk is role-driven, not industry-driven. Truck Drivers and Customer Support Reps stay red across every industry, while Data Analysts and Software Engineers stay green regardless of sector. Industry barely moves the needle.
- Biggest within-role variance: Truck Driver ranges from 58.0% (Technology) to 61.9% (Energy), only a ~4% spread, which is narrow.
- Data Analyst in Manufacturing (34.1%) is the single lowest cell. Truck Driver in Energy (61.9%) is the highest.
- Data Analyst sits consistently in the 34–38% range across all industries, making it one of the more automation-resilient career paths in this dataset.

2.5. Research Question 2: Reskilling Urgency and Skill Gap by Job Role

The second research question examines which roles face the most urgent reskilling need and whether that urgency is matched by a large skill gap. To answer this question, a grouped bar chart comparing two

metrics across 10 job roles, with Job Role on the horizontal axis, Score on the vertical axis, and two bar series, Reskilling Urgency (blue) and Skill Gap Index (orange), displayed side by side for each role.



Key Takeaways:

- Skill Gap Index is consistently higher than Reskilling Urgency across all roles, the orange bar always clears the blue one. This means every role has more skill gap than urgency to act on it.
- Customer Support Rep and Truck Driver lead in Reskilling Urgency (41.3 each), aligning with chart in the Figure 4. Mean Automation Risk by Job Role, where they also had the highest automation risk. High automation risk → high pressure to reskill.
- Skill Gap scores are surprisingly tight across all roles, ranging only from 48.3 to 51.0. This suggests the gap between current and needed skills is roughly uniform regardless of role, the difference is really in how urgently each role needs to close it.
- Data Analyst (31.6) and Software Engineer (30.8) have the lowest Reskilling Urgency, consistent with their lower automation risk. These roles are under less immediate pressure to pivot.

III – SUMMARY

This exploratory data analysis examined automation risk and reskilling urgency across 10 job roles and 8 industries using a synthetic dataset of 14,906 records (post-cleaning). Data quality was strong, with no missing values or duplicates; the only remediation required was removing 94 records with domain-invalid *ai_replacement_score* values exceeding 100.

Regarding the first research question, automation risk is primarily determined by job role rather than industry. Truck Driver (59.7%) and Customer Support Rep (59.0%) carry substantially higher risk than all other roles, driven by the repetitive and rule-based nature of their tasks. A dense middle cluster of six roles (44.7 - 45.4%) shows moderate, roughly equal exposure, while Data Analyst (35.5%) and Software Engineer (34.9%) remain the most automation-resilient paths. The heatmap analysis confirmed that industry context produces only marginal variation within any given role, with within role spreads rarely exceeding 4.5 percentage points.

Regarding the second research question, reskilling urgency mirrors the automation risk pattern: Customer Support Rep and Truck Driver again lead with the highest urgency scores (41.3 each), while Data Analyst (31.6) and Software Engineer (30.8) face the least immediate pressure. A notable cross-role finding is that the Skill Gap Index remains remarkably uniform across all roles (48.3– 51.0), suggesting that workers in every role face a similar magnitude of skill deficit, what differs is how urgently they must close it.

In conclusion, these findings suggest that workforce intervention resources should be prioritized toward high-risk, high-urgency roles such as Truck Driver and Customer Support Rep, where automation pressure is both high and imminent. Meanwhile, roles like Data Analyst and Software Engineer, though not immune to disruption, offer more transition time and may serve as viable reskilling destinations for displaced workers.

Reference

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